Global and Neighbor-Aware Token Learning for Weakly Supervised Remote Sensing Image Semantic Segmentation

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***Abstract*—Image-level weakly supervised remote sensing semantic segmentation aims to learn pixel-level land-cover prediction using only image-level labels, greatly reducing the annotation cost of fully supervised methods. Class activation map (CAM)-based methods are widely used for this task, but they usually focus on the most discriminative regions, leading to incomplete activation and inaccurate boundaries. Recently, vision Transformer (ViT)-based methods have been introduced to alleviate the limitation of CAMs by exploiting token relations and attention mechanisms. However, remote sensing images often contain dense land-cover regions with subtle interclass differences, and patch tokens in deep ViT layers may become oversmoothed without explicit patch-level supervision, weakening local semantic discrimination. Moreover, large intraclass variations and frequent category co-occurrence make image-specific class tokens prone to semantic drift across different remote sensing images. To address these problems, we propose a global and neighbor-aware token learning (GNATL) framework. GNATL contains two complementary modules: neighbor-aware patch token learning (NPTL) and global class token learning (GCTL). NPTL exploits overlapping regions between neighboring crops to construct implicit patch-level constraints, thereby alleviating patch token oversmoothing. Global class token learning (GCTL) dynamically maintains global class tokens as category-level prototypes to guide image-specific class tokens toward stable category semantics. Experiments on the International Society for Photogrammetry and Remote Sensing (ISPRS) Potsdam, ISPRS Vaihingen, and DeepGlobe Land Cover datasets show that GNATL achieves mean Intersection over Union (mIoU) scores of 56.38%, 47.74%, and 62.75%, outperforming the best compared methods by 2.63%, 4.57%, and 2.36%, respectively.**

***Index Terms*—Remote sensing image semantic segmentation, token learning, vision Transformer (ViT), weakly supervised semantic segmentation (WSSS).**

I. Introduction

THE development of remote sensing technology [1], [2], [3] has greatly improved the resolution and availability of aerial and satellite images, providing rich spatial and semantic information [4], [5], [6]. These images record detailed land-cover patterns, object structures, and spatial relationships over large geographic areas, making automatic interpretation increasingly important for large-scale Earth observation. Remote sensing semantic segmentation aims to assign semantic categories to pixels and extract fine-grained land-cover information from remote sensing images. This technique has been widely applied to urban planning, land-cover mapping, ecological monitoring, and disaster assessment [7], [8], [9], [10].

To achieve accurate pixel-level understanding of remote sensing images, fully supervised methods have been widely studied in remote sensing semantic segmentation. These methods have achieved remarkable performance by utilizing dense pixel-level annotations [11], [12], [13]. On the downside, obtaining such annotations for high-resolution remote sensing images is expensive and time-consuming. To resolve this issue, weakly supervised semantic segmentation (WSSS) has attracted attention since it can reduce annotation costs by learning from weak annotations, such as bounding boxes [14], [15], [16], scribbles [17], [18], [19], points [20], and image-level labels [21], [22], [23], [24]. Among these annotations, image-level labels are the easiest to obtain for remote sensing images because they only indicate the presence of land-cover classes, but the lack of explicit spatial information makes image-level WSSS highly challenging.

Most image-level WSSS methods rely on class activation maps (CAMs) [25] to generate localization maps. How-

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ever, CAMs usually activate the most discriminative regions, resulting in incomplete localization maps. To alleviate this problem, many methods have been proposed to improve CAMs from different perspectives, such as region expansion and erasing [26], affinity learning [22], [23], [27], transformation consistency [28], [29], [30], and prototype or contrastive constraints [31], [32]. Although these methods improve the quality of CAMs, their reliance on convolutional classification networks (CNNs) limits the modeling of long-range dependencies, especially in remote sensing scenes.

Recently, vision Transformer (ViT) [33] has achieved remarkable success in various computer vision tasks by modeling images as sequences of patch tokens [34]. Benefiting from self-attention, ViT can capture long-range dependencies and global interactions among image patches, making it suitable for remote sensing images with spatially continuous land-cover regions. As such, ViT has been introduced into image-level WSSS for both natural images and remote sensing images [35], [36], [37], [38], [39], [40], [41]. MCTFormer [35] introduces multiple class tokens into ViT and generates class-specific localization maps through class-to-patch attention, providing a token-based localization framework for WSSS. Following this framework, some methods focus on improving class-token semantics. MCTFormer+ [42] enhances class-token discrimination through contrastive-class-token (CCT) learning, while MoRe [40] identifies the artifact issue in localization attention maps and regularizes class tokens with graph-based category representation and localization-informed constraints. Other methods focus on improving patch token discrimination. ToCo [36] identifies patch token oversmoothing in ViT-based WSSS and introduces global–local token contrast to enhance patch token discrimination. Contrastive token and foreground activation (CTFA) [39] further extends token contrastive learning to weakly supervised remote sensing semantic segmentation and introduces label activation to improve activation maps. In addition, adaptive token linking with segment anything model guidance (ATSG) [41] improves pseudolabel quality by introducing adaptive token linking and segment anything model (SAM)-guided refinement.

In summary, existing methods have improved localization maps by contrastive token learning, patch token regularization, and pseudolabel refinement. However, for remote sensing images, it is nontrivial to preserve both class-level semantic stability and patch-level discriminability, and two major challenges remain unresolved.

1) *How to Maintain Discriminative Patch Tokens for Dense Land-Cover Regions?* Remote sensing images often contain dense land-cover regions with small interclass differences, where adjacent categories may show similar appearances and subtle boundaries. In ViT-based WSSS, patch tokens are mainly optimized by image-level supervision, while self-attention repeatedly aggregates contextual information among patches. As a result, patch tokens may become oversmoothed and lose local discrimination, making it difficult to distinguish adjacent land-cover regions.
2) *How to Achieve Stable Class-Token Under Large Intraclass Variations?* Remote sensing images usually exhibit large intraclass variations, where the same land-cover category may appear with different scales, textures, shapes, and surrounding contexts across images. Under image-level supervision, image-specific class tokens are expected to represent category semantics, but they can be biased by image-specific contexts and co-occurring categories. This may lead to semantic drift of class tokens and degrade the reliability of class-to-patch localization maps.

To address the above challenges, we propose a global and neighbor-aware token learning (GNATL) framework for image-level weakly supervised remote sensing semantic segmentation. Within the proposed global and neighbor-aware token learning (GNATL), we utilize complementary token learning to improve patch-level semantic discrimination and class-level semantic stability. For patch token learning, we design the neighbor-aware patch token learning (NPTL), which can exploit the overlapping regions between neighboring crops to construct implicit patch-level constraints. Since overlapping regions correspond to the same geographic areas, neighbor-aware patch token learning (NPTL) enhances patch-level supervision without pixel-level annotations, alleviating patch token semantic homogenization and preserving local discrimination. For class-token learning, we design the GCTL to dynamically maintain global class tokens as category-level semantic prototypes. These global prototypes provide stable semantic guidance for image-specific class tokens, reducing semantic drift.

The main contributions of this article are summarized as follows.

1) We propose NPTL to preserve discriminative patch tokens for dense land-cover regions. By exploiting the overlapping regions between neighboring crops, NPTL constructs implicit patch-level constraints to alleviate patch token oversmoothing and preserve local semantic discrimination.
2) We propose GCTL to maintain stable class-token semantics under large intraclass variations. GCTL dynamically maintains global class tokens as category-level semantic prototypes and provides stable semantic guidance for image-specific class tokens, reducing semantic drift caused by image-specific contexts and category co-occurrence.
3) Experiments on the International Society for Photogrammetry and Remote Sensing (ISPRS) Potsdam, ISPRS Vaihingen, and DeepGlobe Land Cover datasets demonstrate the effectiveness and generalization ability of the proposed method. GNATL achieves mean Intersection over Union (mIoU) scores of 56.38%, 47.74%, and 62.75% on the three datasets, respectively, outperforming the best compared methods by 2.63%, 4.57%, and 2.36%.

The remainder of this article is organized as follows. Section II reviews related work on image-level WSSS and WSSS for remote sensing. Section III introduces the proposed GNATL. Section IV presents comparative experiments, token-level analyses, parameter studies, and efficiency analysis. Finally, Section V concludes this article.

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II. Related Work

A. Image-Level WSSS

Image-level WSSS usually starts from CAMs [25], which provide coarse object seeds from image-level labels. However, CAMs generated by convolutional classifiers tend to highlight only the most discriminative regions and therefore require additional refinement or regularization to recover complete object regions. Early WSSS methods commonly refine CAM seeds by modeling pixel relations. AffinityNet [22] learns semantic affinities from reliable seed regions, IRNet [23] further exploits interpixel relations for boundary-aware pseudo masks, and iterative affinity learning [27] constructs affinity matrices from learned high-level representations. Other methods improve localization by enforcing transformation consistency, such as SEAM [28], PuzzleCAM [29], and crop-consistency learning [30]. More recent studies further exploit prototype-based contrastive learning and progressive region expansion to mine more reliable pixel-level supervision from image-level labels [26], [31], [32]. Wu et al. [43] model class-wise label distributions with an adaptive Gaussian mixture framework and exploit semantic correlations and cross-label constraints to improve pseudolabel reliability under different forms of weak supervision.

With the development of ViTs [33], token-based representations have provided a new direction for image-level WSSS. ViTs explicitly model long-range dependencies through self-attention, making class-to-patch and patch-to-patch relations useful for object localization. MCTFormer [35] introduces multiple class tokens and generates class-specific localization maps through class-to-patch attention. MCTFormer+ [42] further strengthens class-token discrimination and uses patch-level affinity from Transformer attention to refine localization maps. WeakTr [37] explores a plain ViT for WSSS and adaptively aggregates self-attention maps from different heads to improve localization. MoRe [40] observes that class-patch attention may activate irrelevant patch regions and regularizes such attention with graph-based category representations and localization-informed constraints.

Closely related to ViT-based token learning, recent contrastive language-image pretraining (CLIP)-based WSSS methods further introduce vision–language priors into dense localization. CLIMS [44] exploits language-guided image matching to enhance class-discriminative activation, while CLIP-ES [45] uses CLIP [46] to generate text-driven localization cues for weakly supervised segmentation. WeCLIP [38] shows that a frozen CLIP backbone can serve as a strong semantic representation for WSSS; WeCLIP+ [47] further combines frozen CLIP and self-DIstillation with NO labels (DINO) features to strengthen semantic representation and online pseudolabel refinement, and ExCEL [48] explores CLIP's dense knowledge through patch-text alignment rather than only global image–text matching.

B. WSSS for Remote Sensing

Remote sensing semantic segmentation has been widely studied because it supports applications such as urban analysis, environmental monitoring, and land-cover mapping. Fully supervised methods based on FCNs [11], U-Net [12], and DeepLab [13] have achieved strong performance, and many remote sensing segmentation models further enhance multi-scale context modeling and boundary perception. However, high-resolution remote sensing images require labor-intensive pixel-level annotations, which makes weakly supervised learning attractive. Compared with natural images, remote sensing images contain more complex spatial layouts, large intra-class variations, subtle interclass differences, and many small objects. These properties make pseudolabel generation more difficult and reduce the reliability of CAM-based seeds.

Several studies have explored WSSS specifically for remote sensing scenes. FlipCAM [49] designs a feature-level flipping augmentation strategy for weakly supervised building extraction, improving CAM consistency under horizontal transformations. CocoaNet [50] introduces a consistency-constrained multiclass attention mechanism and global–local adaptive attention to improve remote sensing WSSS. CTFA [39] adopts a ViT-based framework for remote sensing WSSS and uses contrastive token learning to alleviate token over-smoothing, while a dual-branch decoder activates foreground features in pseudolabels. PLNS [51] suppresses pseudolabel noise for remote sensing WSSS, and ATSG [41] further exploits adaptive token linking and Segment Anything Model guidance to improve pseudolabel quality and boundary completeness in complex remote sensing scenes. Recent text-driven remote sensing WSSS also explores semantic priors from vision–language models [52]. SGCT-Net [53] adopts two nonshared student networks for cross-teaching and combines dynamic thresholding, noise filtering, and Segment Anything Model guidance to generate higher quality CAMs in high-resolution remote sensing imagery. SuperScribble [54] introduces superpixel- and pixel-level constraints for remote sensing semantic segmentation with scribble annotations.

Despite recent progress, weakly supervised remote sensing semantic segmentation remains challenging. Existing methods often overlook the spatial continuity between neighboring crops, even though such crops naturally contain overlapped regions that can provide implicit patch-level constraints. Moreover, class-token representations are usually learned independently for each image and lack stable category-level References, which may cause inconsistent responses across images with diverse appearances. To address these limitations, our method introduces NPTL to exploit overlapped neighboring crops for patch token consistency, and GCTL to maintain dynamically updated global class tokens as category-level semantic prototypes.

III. Methodology

In this section, we present the proposed GNATL framework for image-level weakly supervised remote sensing semantic segmentation. GNATL is designed to alleviate patch token oversmoothing through neighbor-aware patch consistency and reduce semantic drift of image-specific class tokens through global class-token guidance. In particular, we first introduce the overall framework of GNATL. Then, we describe NPTL, which exploits the spatial overlap between neighboring crops

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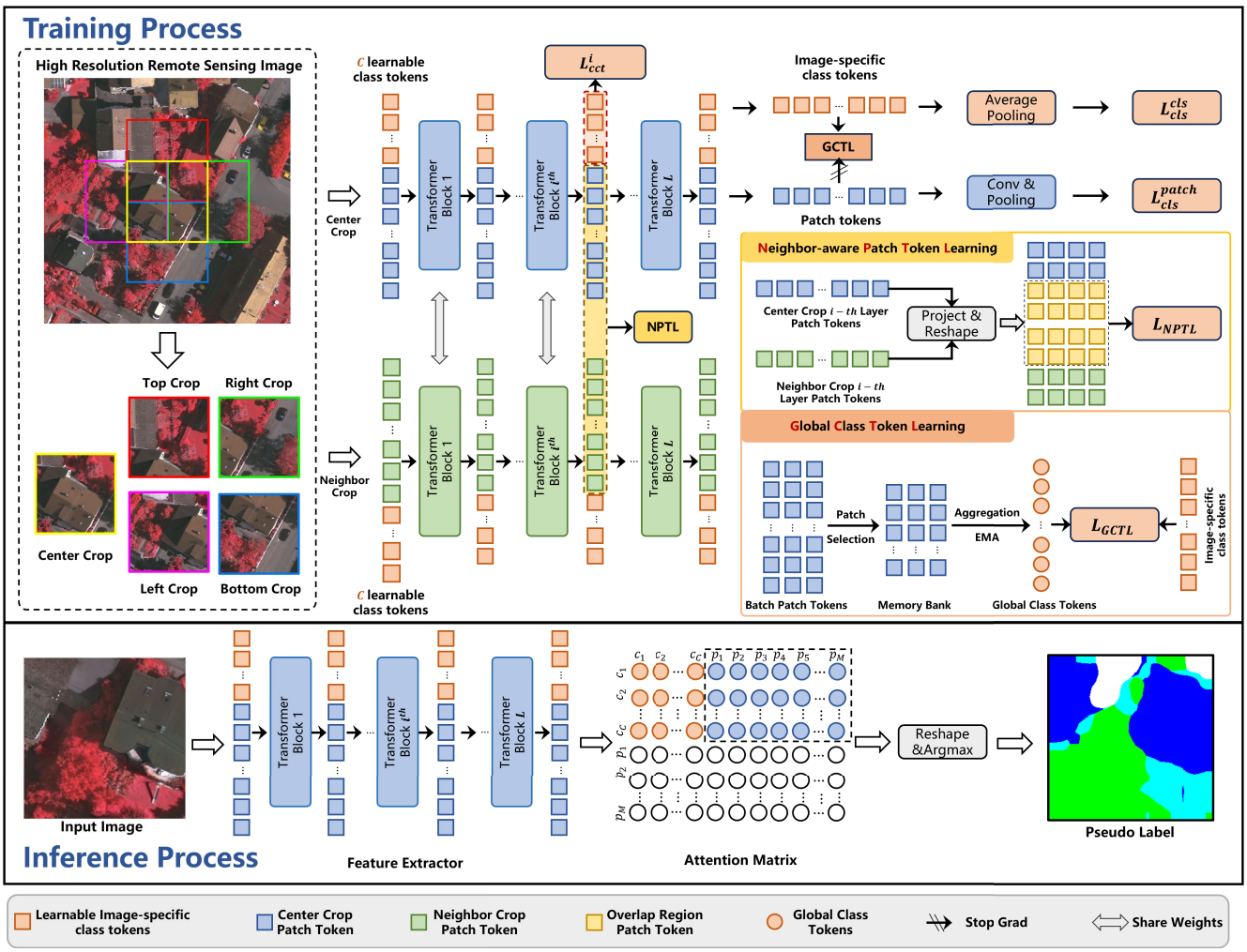


Fig. 1. Overall framework of the proposed GNATL. During training, a center crop and one neighboring crop from a high-resolution remote sensing image are fed into shared-weight ViT branches. GNATL improves token learning through two complementary strategies: NPTL, which enforces consistency on overlapped regions between neighboring crops, and GCTL, which maintains global class tokens to guide image-specific class tokens. During inference, a single image is fed into the trained network, and its class-to-patch attention is reshaped into class-specific localization maps and converted into a pseudolabel by pixel-wise argmax.

to provide implicit patch-level supervision. Next, we introduce GCTL, which maintains dynamically updated global class tokens as category-level semantic references to stabilize image-specific class tokens. Finally, we formulate the overall training objective.

A. Overall Framework of GNATL

Image-level supervision only indicates the presence or absence of land-cover categories but does not provide explicit pixel-level location information. In ViT-based weakly supervised remote sensing semantic segmentation, such limited supervision mainly affects two types of tokens. First, patch tokens are updated through self-attention by aggregating contextual information from other tokens. With the repeated token mixing in deep Transformer layers, patch representations may gradually become homogeneous, especially in the absence of explicit patch-level constraints. This leads to patch token oversmoothing and weakens local semantic discrimination. Second, image-specific class tokens are easily affected by complex local contexts and category co-occurrence in remote sensing scenes. As a result, they may suffer from semantic drift and fail to represent stable category semantics.

To address these issues, we propose GNATL, which enhances token learning from both patch token and class-token perspectives. As shown in Fig. 1, GNATL is built upon a shared-weight ViT-based CNN. Given a high-resolution remote sensing image, we first generate a set of image crops using a sliding-window strategy. For each crop, one neighboring crop is randomly sampled from its spatial neighborhood, and the paired crops are fed into the shared CNN.

For patch token learning, the proposed NPTL (as shown in Algorithm 1) exploits the overlapped regions between neighboring crops to constrain deep patch tokens. Since these regions correspond to the same image area but are encoded under different contexts, NPTL enforces semantic consistency between them, thereby providing implicit patch-level supervision and alleviating patch token oversmoothing.

For class-token learning, the proposed GCTL (as shown in Algorithm 2) maintains dynamic global class tokens as semantic prototypes. GCTL selects and aggregates high-response patch tokens according to class-to-patch attention to update the global prototype of each category. These global prototypes

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Algorithm 1 Neighbor-Aware Patch Token Learning

Input: crop x_i, neighboring crop x_j, classification network f_θ, auxiliary patch projector g_ϕ, selected deep layers $\mathcal{S}$.
Output: Neighbor-aware patch token learning loss $\mathcal{L}_{\text{NPTL}}$.

Feed x_i and x_j into the shared classification network f_θ.
Extract multi-layer patch tokens from x_i and x_j.
Initialize $\mathcal{L}_{\text{NPTL}} = 0$.
for each selected layer $l \in \mathcal{S}$ **do**
  Extract overlapped patch tokens $\mathbf{P}^l_{i,\Omega}$.
  Extract overlapped patch tokens $\mathbf{P}^l_{j,\Omega}$.
  Project patch tokens of x_i:

$$\mathbf{H}^l_i = g_\phi(\mathbf{P}^l_i).$$

  Project patch tokens of x_j:

$$\mathbf{H}^l_j = g_\phi(\mathbf{P}^l_j).$$

  Convert projected representations of the overlapped regions into probability distributions.
  Compute layer-wise neighbor-aware consistency loss:

$$\mathcal{L}^l_{\text{NPTL}} = \text{JS}(\mathbf{Q}^l_{i,\Omega}, \mathbf{Q}^l_{j,\Omega}).$$

  Update:

$$\mathcal{L}_{\text{NPTL}} \leftarrow \mathcal{L}_{\text{NPTL}} + \mathcal{L}^l_{\text{NPTL}}.$$

end for
return $\mathcal{L}_{\text{NPTL}}$.

Algorithm 2 Global Class Token Learning

Input: class tokens $\mathbf{T}_i$, patch tokens $\mathbf{P}^L_i$, class-to-patch attention $\mathbf{A}_i$, image-level labels $\mathbf{y}_i$, global prototypes $\mathbf{G}$.
Output: Global class token learning loss $\mathcal{L}_{\text{GCTL}}$ and updated global prototypes $\mathbf{G}$.

Initialize batch token memory banks $\{\mathcal{M}^c_{\mathcal{B}}\}^C_{c=1}$.
for each sample $x_i \in \mathcal{B}$ **do**
  Select high-response patch $\mathcal{K}_{i,c}$.
  Compute normalized weights $\alpha_{i,c,n}$ for $n \in \mathcal{K}_{i,c}$.
  Store selected patch tokens and weights into $\mathcal{M}^c_{\mathcal{B}}$.
end for
for each category $c = 1, \ldots, C$ **do**
  Aggregate $\mathcal{M}^c_{\mathcal{B}}$ to obtain the batch-level prototype $\bar{\mathbf{m}}_c$.
  if $\mathbf{g}_c$ is not initialized **then**
    Initialize $\mathbf{g}_c$ using $\bar{\mathbf{m}}_c$.
  else
    Update $\mathbf{g}_c$ by exponential moving average.
  end if
end for
Compute $\mathcal{L}_{\text{GCTL}}$ by aligning positive image-specific class tokens with stop-gradient global prototypes.
return $\mathcal{L}_{\text{GCTL}}$ and $\mathbf{G}$.

then guide image-specific class tokens, thereby alleviating semantic drift.

With the proposed NPTL and GCTL modules, GNATL learns more discriminative patch tokens and more stable class tokens, enabling better performance in image-level weakly supervised remote sensing semantic segmentation.

During inference, a single image is fed into the trained CNN to obtain a class-to-patch attention matrix $\mathbf{A} \in \mathbb{R}^{C\times M}$, where C denotes the number of semantic categories and M denotes the number of patch tokens. The attention vector for each category is reshaped to its spatial form to produce a class-specific localization map. A pixel-wise argmax operation across the category dimension is then applied to generate the final pseudolabel, as illustrated in the lower branch of Fig. 1.

B. Neighbor-Aware Patch Token Learning

To alleviate patch token oversmoothing, we propose NPTL, which constructs implicit patch-level supervision from the overlapped regions between neighboring crops. Instead of relying on foreground–background constraints, NPTL directly exploits the spatial correspondence naturally formed by neighboring crops. Since the overlapped regions correspond to the same image area but are encoded under different surrounding contexts, their patch representations are encouraged to be consistent. This neighbor-aware consistency constrains deep patch tokens and preserves their local semantic discrimination.

Given a high-resolution remote sensing image I, we first divide it into a set of crops using a sliding-window strategy

$$\mathcal{X} = \{x_i\}^M_{i=1}. \tag{1}$$

For each crop x_i, one neighboring crop x_j is randomly sampled from its four-neighborhood $\mathcal{N}(i)$, where $\mathcal{N}(i)$ contains the upper, lower, left, and right neighboring crops of x_i. The two crops are then fed into the shared CNN f_θ to extract multilayer patch tokens

$$\{\mathbf{P}^l_i\}^L_{l=1} = f^p_\theta(x_i) \tag{2}$$

$$\{\mathbf{P}^l_j\}^L_{l=1} = f^p_\theta(x_j) \tag{3}$$

where $\mathbf{P}^l_i \in \mathbb{R}^{N\times D}$ and $\mathbf{P}^l_j \in \mathbb{R}^{N\times D}$ denote the patch tokens at the lth Transformer block, N is the number of patch tokens, and D is the token dimension.

Since adjacent crops are sampled with a fixed spatial overlap, x_i and x_j share half of their patch tokens in the overlapped region. At the lth Transformer block, we denote the patch tokens corresponding to the overlapped region in x_i and x_j as $\mathbf{P}^l_{i,\Omega}$ and $\mathbf{P}^l_{j,\Omega}$, respectively,

$$\mathbf{P}^l_{i,\Omega} = \mathbf{P}^l_i[\Omega_i] \tag{4}$$

$$\mathbf{P}^l_{j,\Omega} = \mathbf{P}^l_j[\Omega_j] \tag{5}$$

where Ω_i and Ω_j denote the overlapped patch token sets in x_i and x_j, and $|\Omega_i| = |\Omega_j| = N/2$.

To measure the consistency between overlapped patch tokens, we introduce an auxiliary patch projector $g_\phi(\cdot)$, which transforms patch tokens into a compact projection space

$$g_\phi : \mathbb{R}^D \to \mathbb{R}^{D'} \tag{6}$$

where D' denotes the dimension of the projected representation. Since patch token oversmoothing mainly appears in deep Transformer layers, the auxiliary patch projector is applied to a set of selected deep layers

$$\mathcal{S} = \{L - K + 1, \ldots, L\}. \tag{7}$$

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For each selected layer $l \in \mathcal{S}$, the patch tokens of x_i and x_j are projected as follows:

$$\mathbf{H}_i^l = g_\phi\left(\mathbf{P}_i^l\right) \tag{8}$$

$$\mathbf{H}_j^l = g_\phi\left(\mathbf{P}_j^l\right) \tag{9}$$

where $\mathbf{H}_i^l \in \mathbb{R}^{N\times D'}$ and $\mathbf{H}_j^l \in \mathbb{R}^{N\times D'}$ denote the projected patch representations.

To compute the consistency between overlapped patches, we first extract the projected representations of the overlapped regions and convert them into probability distributions over the projection dimension

$$\mathbf{Q}_{i,\Omega}^l = \text{softmax}\left(\frac{\mathbf{H}_i^l\left[\Omega_i\right]}{\tau}\right) \tag{10}$$

$$\mathbf{Q}_{j,\Omega}^l = \text{softmax}\left(\frac{\mathbf{H}_j^l\left[\Omega_j\right]}{\tau}\right) \tag{11}$$

where τ is a temperature coefficient, and the softmax operation is performed along the projection dimension. The layer-wise neighbor-aware consistency loss is then defined as follows:

$$\mathcal{L}_{\text{NPTL}}^l = \text{JS}\left(\mathbf{Q}_{i,\Omega}^l, \mathbf{Q}_{j,\Omega}^l\right) \tag{12}$$

where $\text{JS}(\cdot,\cdot)$ denotes the Jensen–Shannon divergence, which is computed between the aligned overlapped regions and averaged over the overlapped patch dimension. The final NPTL loss is obtained by aggregating the consistency loss over all selected deep layers

$$\mathcal{L}_{\text{NPTL}} = \sum_{l\in\mathcal{S}} \mathcal{L}_{\text{NPTL}}^l. \tag{13}$$

C. Global Class Token Learning

To alleviate the semantic drift of image-specific class tokens, we propose GCTL. GCTL maintains dynamically updated global class tokens as category-level semantic prototypes. Instead of relying only on image-level classification supervision to optimize class tokens, GCTL selects high-response patch tokens according to class-to-patch attention and stores them in a batch-level token memory bank. After each minibatch is processed, the stored patch tokens are aggregated into batch-level prototypes, which are then used to update the global semantic prototypes by exponential moving average. The updated global prototypes provide stable category-level guidance for image-specific class tokens.

For a crop x_i, we denote its last-layer class tokens as follows:

$$\mathbf{T}_i = \left\{\mathbf{t}_{i,c}\right\}_{c=1}^{C} \in \mathbb{R}^{C\times D} \tag{14}$$

and its last-layer patch tokens as follows:

$$\mathbf{P}_i^L = \left\{\mathbf{p}_{i,n}\right\}_{n=1}^{N} \in \mathbb{R}^{N\times D} \tag{15}$$

where C is the number of categories, N is the number of patch tokens, and D is the token dimension. We further denote the last-layer class-to-patch attention as follows:

$$\mathbf{A}_i \in \mathbb{R}^{C\times N} \tag{16}$$

where $\mathbf{A}_{i,c,n}$ indicates the attention response between the cth class token and the nth patch token.

For each crop x_i and category c, GCTL first selects the top-k_g high-response patch tokens according to the class-to-patch attention

$$\mathcal{K}_{i,c} = \text{TopK}\left(\mathbf{A}_{i,c}, k_g\right) \tag{17}$$

where $\mathcal{K}_{i,c}$ denotes the index set of the selected high-response patch tokens for category c. The corresponding attention weights are normalized as follows:

$$\alpha_{i,c,n} = \frac{\mathbf{A}_{i,c,n}}{\sum_{r\in\mathcal{K}_{i,c}} \mathbf{A}_{i,c,r} + \epsilon} \tag{18}$$

where $n \in \mathcal{K}_{i,c}$ and ϵ are a small constant for numerical stability.

For each minibatch $\mathcal{B}$, GCTL constructs a batch-level token memory bank for each category by collecting high-response patch tokens from positive samples. For category c, the memory bank is defined as follows:

$$\mathcal{M}_{\mathcal{B}}^c = \left\{\left(\mathbf{p}_{i,n}, \alpha_{i,c,n}\right) \mid y_{i,c} = 1,\ x_i \in \mathcal{B},\ n \in \mathcal{K}_{i,c}\right\}. \tag{19}$$

Here, only samples with positive image-level labels are used because image-level annotations provide reliable category presence information.

After all samples in the minibatch are processed, the stored high-response patch tokens are aggregated to form the batch-level prototype of category c. We first define the number of positive samples for category c in the minibatch as follows:

$$N_{\mathcal{B}}^c = \sum_{x_i\in\mathcal{B}} y_{i,c}. \tag{20}$$

If $N_{\mathcal{B}}^c > 0$, the batch-level prototype is computed as follows:

$$\bar{\mathbf{m}}_c = \frac{1}{N_{\mathcal{B}}^c} \sum_{x_i\in\mathcal{B}} y_{i,c} \sum_{n\in\mathcal{K}_{i,c}} \alpha_{i,c,n}\mathbf{p}_{i,n}. \tag{21}$$

In this way, $\bar{\mathbf{m}}_c$ summarizes the high-response patch tokens of category c within the current minibatch.

We maintain a set of global class tokens as category-level semantic prototypes

$$\mathbf{G} = \{\mathbf{g}_c\}_{c=1}^{C} \in \mathbb{R}^{C\times D} \tag{22}$$

where $\mathbf{g}_c$ denotes the global prototype of category c.

The global prototype is updated by exponential moving average

$$\mathbf{g}_c \leftarrow \mu\mathbf{g}_c + (1-\mu)\,\bar{\mathbf{m}}_c \tag{23}$$

where μ is the momentum coefficient.

After updating the global prototypes, GCTL uses them to guide image-specific class tokens. For each positive category, the image-specific class token $\mathbf{t}_{i,c}$ is encouraged to approach the corresponding global prototype $\mathbf{g}_c$. The GCTL loss is defined as follows:

$$\mathcal{L}_{\text{GCTL}} = \frac{\sum_{i=1}^{B}\sum_{c=1}^{C} y_{i,c}\left[1 - \cos\left(\mathbf{t}_{i,c}, \mathbf{g}_c\right)\right]}{\sum_{i=1}^{B}\sum_{c=1}^{C} y_{i,c} + \epsilon} \tag{24}$$

where $\cos(\cdot,\cdot)$ denotes cosine similarity. The alignment is applied only to positive categories because image-level labels provide reliable category presence information but do not provide reliable absence localization. Through this dynamic prototype guidance, GCTL alleviates the semantic drift of image-specific class tokens and improves the stability of category representations.

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D. Objective Functions

The overall training objective consists of the base classification loss, the NPTL loss, and the GCTL loss. The base loss includes the class-token classification loss, the patch token classification loss, and the class-token contrastive constraint loss

$$\mathcal{L}_{\text{base}} = \mathcal{L}_{\text{cls}}^{\text{cls}} + \lambda_{\text{patch}}\mathcal{L}_{\text{cls}}^{\text{patch}} + \lambda_{\text{cct}}\mathcal{L}_{\text{cct}} \tag{25}$$

where $\mathcal{L}_{\text{cls}}^{\text{cls}}$ denotes the image-level multilabel classification loss computed from class tokens, $\mathcal{L}_{\text{cls}}^{\text{patch}}$ denotes the image-level multilabel classification loss computed from patch tokens, and $\mathcal{L}_{\text{cct}}$ denotes the class-token contrastive constraint loss. The CCT loss follows the setting of MCTFormer. The coefficients λ_{patch} and λ_{cct} control the contributions of $\mathcal{L}_{\text{cls}}^{\text{patch}}$ and $\mathcal{L}_{\text{cct}}$, respectively. The proposed NPTL and GCTL losses are incorporated as token-level regularization terms. The total training objective is formulated as follows:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{base}} + \lambda_{\text{nptl}}\mathcal{L}_{\text{NPTL}} + \lambda_{\text{gctl}}\mathcal{L}_{\text{GCTL}} \tag{26}$$

where λ_{nptl} and λ_{gctl} control the contributions of $\mathcal{L}_{\text{NPTL}}$ and $\mathcal{L}_{\text{GCTL}}$, respectively.

IV. Experiments

A. Experimental Setup

1) Datasets: In this section, we conduct experiments on the ISPRS Potsdam, ISPRS Vaihingen, and DeepGlobe Land Cover datasets to evaluate the effectiveness and generalization ability of the proposed GNATL framework for image-level weakly supervised remote sensing semantic segmentation.

We first introduce the experimental settings, including datasets, evaluation metrics, and implementation details. Then, we compare the proposed method with existing WSSS methods. Finally, we conduct ablation studies to verify the effectiveness of the proposed NPTL and GCTL modules.

To evaluate the effectiveness of the proposed GNATL framework under different remote sensing scenes, we conduct experiments on the ISPRS 2-D Semantic Labeling Benchmark [55], including the Potsdam and Vaihingen datasets, and the DeepGlobe Land Cover dataset [56]. The two ISPRS datasets contain very high-resolution aerial images of urban areas with diverse building structures, vegetation distributions, road layouts, and object scales, whereas DeepGlobe contains large-scale satellite imagery from more diverse geographical environments.

The Potsdam dataset consists of 38 aerial image tiles, each with a size of 6000 × 6000 pixels and a ground sampling distance of 5 cm. Each image is provided with four spectral bands, including infrared, red, green, and blue channels.

The Vaihingen dataset contains 33 aerial image tiles with different spatial sizes and a ground sampling distance of 9 cm. Both datasets provide pixel-level annotations for six semantic categories, including impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background.

The DeepGlobe Land Cover dataset was released in 2018 and contains large-scale satellite images with seven original categories: urban, agriculture, rangeland, forest, water, barren, and unknown. Because unknown accounts for only a negligible proportion of the dataset, it is excluded from our experiments, and the remaining six land-cover categories are used for quantitative evaluation. Compared with the two urban aerial datasets, DeepGlobe presents broader land-cover distributions and greater intraclass variation, providing a complementary and more challenging evaluation setting.

For Potsdam and Vaihingen, the original high-resolution images are cropped into image patches with a size of 256×256 and a stride of 128. For DeepGlobe, the images are cropped into 512 × 512 patches with a stride of 256. Both settings produce a 50% overlap between adjacent crops. The image-level label of each cropped patch is generated according to the semantic categories appearing in the corresponding ground-truth mask. In particular, if a category appears in a cropped patch, the corresponding image-level label is set to 1; otherwise, it is set to 0.

2) Evaluation Metrics: We adopt mIoU as the evaluation metric to assess pseudolabel quality. All class-wise Intersection over Union (IoU) and mIoU values are reported as percentages. For each category, the IoU is computed as follows:

$$\text{IoU}_c = \frac{TP_c}{TP_c + FP_c + FN_c} \tag{27}$$

where TP_c, FP_c, and FN_c denote the numbers of true-positive, false-positive, and false-negative pixels for category c, respectively. The final mIoU is obtained by averaging the IoU values over all categories

$$\text{mIoU} = \frac{1}{C}\sum_{c=1}^{C}\text{IoU}_c \tag{28}$$

where C denotes the number of categories.

3) Implementation Details: The proposed GNATL is implemented based on DeiT-Small. The CNN is trained for 50 epochs with a batch size of 32. We use the AdamW optimizer with an initial learning rate of 1.25×10^{-5} and a weight decay of 0.05. A cosine learning rate scheduler is adopted.

For the training objectives, the number of class-token contrastive constraint layers is set to 6. For NPTL, we select the last 6 Transformer layers to compute the neighbor-aware patch consistency loss, and the loss weight is set to 0.5. One neighboring crop is randomly sampled from the top, bottom, left, or right spatial direction. For GCTL, an 8-epoch warm-up is used before activating the prototype guidance, and the top-5 high-response patch tokens are selected to update the global class prototypes. The momentum coefficient for global prototype updating is set to 0.8, and the GCTL loss weight is set to 0.1.

After classification training, the generated activation maps are converted into pseudolabels. All quantitative results are directly calculated from these pseudolabels without conditional random field (CRF) postprocessing or an additional downstream semantic segmentation network. This protocol avoids introducing variables from a separate segmentation architecture, optimization strategy, or postprocessing configuration and directly evaluates the intrinsic quality of the pseudolabels.

All experiments are conducted on a Linux server equipped with two NVIDIA GeForce RTX 3090 GPUs. The

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TABLE I
COMPARISON OF PSEUDOLABEL QUALITY WITH OTHER WSSS METHODS ON THE POTSDAM DATASET

Method	Imp. Surf. (%)	Building (%)	Low Veg. (%)	Tree (%)	Car (%)	Clutter (%)	mIoU (%)
MCTFormer+ [42]$_{TPAMI'24}$	61.60	71.09	50.72	55.46	42.60	28.08	51.59
CTFA [39]$_{TGRS'24}$	46.96	57.48	**60.73**	59.09	57.90	16.83	49.83
MoRe [40]$_{AAAI'25}$	64.07	75.30	59.69	41.68	59.34	18.96	53.17
WeCLIP+ [47]$_{TPAMI'25}$	57.54	47.96	48.63	43.82	**64.77**	4.40	44.52
WeakTr [37]$_{TIP'26}$	55.13	37.08	38.81	38.61	21.48	43.92	39.17
ATSG [41]$_{TGRS'26}$	**67.76**	59.48	54.58	**62.61**	19.60	**58.49**	53.75
Ours	66.26	**77.69**	60.48	57.55	42.48	33.80	**56.38**

experimental environment includes PyTorch 2.0.1, Compute Unified Device Architecture (CUDA) 11.7, and Python 3.9.2.

B. Comparison With State-of-the-Art Methods

To evaluate the effectiveness of the proposed GNATL, we compare it with several representative WSSS methods, including MCTFormer+ [42], CTFA [39], MoRe [40], WeCLIP+ [47], ATSG [41], and WeakTr [37]. These methods are selected because they are closely related to Transformer-based activation map generation, class-token learning, and token-level regularization.

MCTFormer+ [42] introduces multiple class tokens into ViTs and generates class-specific activation maps through class-to-patch attention. CTFA [39] improves ViT-based weakly supervised remote sensing segmentation by introducing contrastive token learning and label activation. MoRe [40] regularizes class-patch attention by introducing graph category representation and localization-informed regularization. WeCLIP+ [47] combines frozen CLIP and DINO backbones with strengthened online pseudolabel refinement, providing a recent foundation-model-based comparison. ATSG [41] introduces adaptive token linking and incorporates SAM guidance to improve pseudolabel generation. WeakTr [37] explores plain ViTs and adaptively fuses multilayer and multihead self-attention maps for activation map generation.

Compared with these methods, GNATL focuses on two token learning problems in image-level weakly supervised remote sensing semantic segmentation. In particular, NPTL constrains deep patch tokens through neighbor-aware consistency, while GCTL maintains global class prototypes to guide image-specific class tokens. For fair comparison, all methods are evaluated under the same image-level weakly supervised setting, and the reported results are obtained without CRF postprocessing.

1) Results on the Potsdam Dataset: Table I reports the quantitative comparison on the Potsdam dataset. Compared with existing WSSS methods, the proposed GNATL achieves the best overall performance, with an mIoU of 56.38%. It outperforms the strongest competing method, ATSG, by 2.63% mIoU, demonstrating the effectiveness of jointly enhancing patch token and class-token learning.

In terms of category-wise performance, GNATL achieves the best IoU on the building category and obtains competitive results on impervious surfaces and low vegetation. In particular, GNATL improves the building IoU to 77.69%, outperforming MoRe by 2.39% and MCTFormer+ by 6.60%. For impervious surfaces and low vegetation, GNATL also achieves IoU values of 66.26% and 60.48%, respectively. These results indicate that GNATL can produce more reliable segmentation results for large-area and spatially continuous land-cover categories.

Fig. 2 further presents visualization results of the WSSS methods on the Potsdam dataset. Compared with other representative methods, GNATL generates more complete and spatially coherent segmentation maps. For building and impervious surface regions, the proposed method better preserves the structural integrity of large objects and reduces fragmented predictions. For vegetation-related categories, GNATL also alleviates category confusion between low vegetation and trees.

2) Results on the Vaihingen Dataset: Table II reports the quantitative comparison on the Vaihingen dataset. The proposed GNATL achieves the best overall performance, with an mIoU of 47.74%. Compared with the strongest competing method, MoRe, GNATL improves the mIoU by 4.57%. This result further demonstrates the effectiveness of the proposed token learning strategy on remote sensing scenes with complex spatial layouts and diverse land-cover distributions.

For category-wise performance, GNATL achieves the best IoU on the building category, reaching 69.01%. While some competing methods obtain higher IoU on specific categories, such as ATSG on impervious surfaces, low vegetation, and clutter, and MoRe on trees and cars, their performance is less balanced across all categories. On the contrary, GNATL achieves the highest overall mIoU of 47.74%, demonstrating more stable performance across different land-cover categories.

Fig. 3 shows the visualization results of the WSSS methods on the Vaihingen dataset. Compared with other representative methods, GNATL generates more coherent segmentation regions and better preserves the structures of buildings and impervious surfaces. For vegetation-related areas, GNATL also produces more stable predictions and reduces fragmented responses. These visualization results are consistent with the quantitative comparison and further show that NPTL and GCTL can improve token representations under complex remote sensing scenes.

3) Results on the DeepGlobe Dataset: Table III reports the pseudolabel comparison on the DeepGlobe dataset. GNATL achieves the best overall mIoU of 62.75%, outperforming

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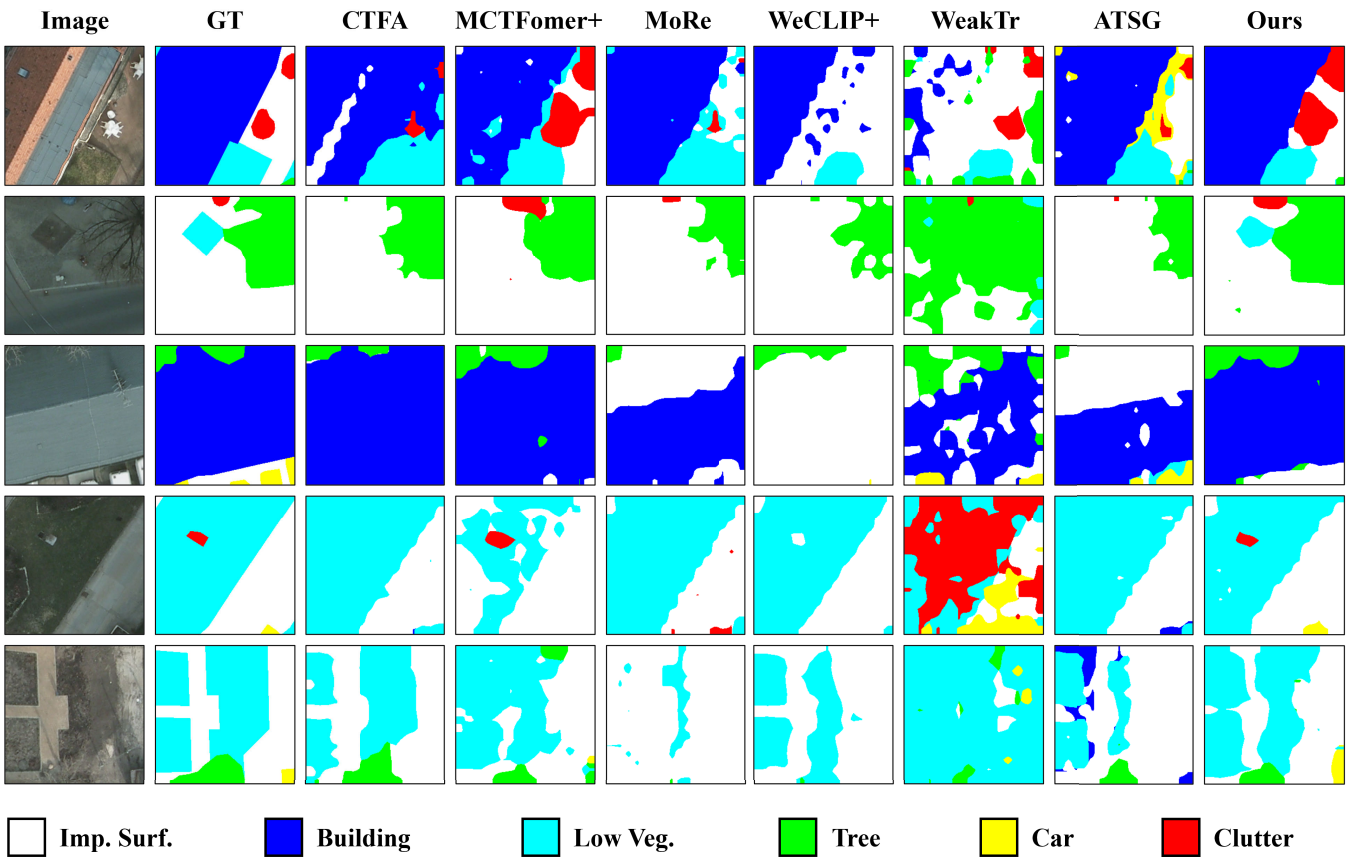


Fig. 2. Qualitative comparison of pseudolabels generated by different WSSS methods on the Potsdam dataset.

TABLE II
COMPARISON OF PSEUDOLABEL QUALITY WITH OTHER WSSS METHODS ON THE VAIHINGEN DATASET

Method	Imp. Surf. (%)	Building (%)	Low Veg. (%)	Tree (%)	Car (%)	Clutter (%)	mIoU (%)
MCTFormer+ [42]$_{TPAMI'24}$	53.94	58.64	39.07	57.91	21.78	26.48	42.97
CTFA [39]$_{TGRS'24}$	47.00	45.85	27.47	62.09	40.98	11.91	39.22
MoRe [40]$_{AAAI'25}$	50.48	56.33	20.35	**64.67**	**41.23**	25.97	43.17
WeCLIP+ [47]$_{TPAMI'25}$	54.38	56.02	30.59	21.03	40.41	1.32	33.96
WeakTr [37]$_{TIP'26}$	49.28	30.31	47.71	7.97	23.92	27.43	31.11
ATSG [41]$_{TGRS'26}$	**64.80**	23.50	**59.54**	43.90	10.96	**53.01**	42.62
Ours	59.70	**69.01**	46.65	57.84	27.24	25.98	**47.74**

TABLE III
COMPARISON OF PSEUDOLABEL QUALITY WITH OTHER WSSS METHODS ON THE DEEPGLOBE DATASET

Method	Urban (%)	Agriculture (%)	Rangeland (%)	Forest (%)	Water (%)	Barren (%)	mIoU (%)
MCTFormer+ [42]$_{TPAMI'24}$	60.48	83.22	24.82	75.81	67.73	48.80	60.14
CTFA [39]$_{TGRS'24}$	60.56	82.12	19.78	78.84	70.13	48.18	59.94
MoRe [40]$_{AAAI'25}$	**60.83**	79.93	14.42	67.27	56.23	42.15	53.47
WeCLIP+ [47]$_{TPAMI'25}$	47.12	74.99	8.27	74.08	**71.85**	43.18	53.25
WeakTr [37]$_{TIP'26}$	23.78	45.59	20.15	64.47	38.07	45.38	39.57
ATSG [41]$_{TGRS'26}$	58.26	83.71	23.37	76.44	69.58	50.99	60.39
Ours	55.90	**84.87**	**36.97**	**79.51**	65.48	**53.76**	**62.75**

the strongest competing method, ATSG, by 2.36%. It obtains the highest IoU for Agriculture, Rangeland, Forest, and Barren, demonstrating balanced performance across large-scale satellite scenes. The improvement is particularly evident

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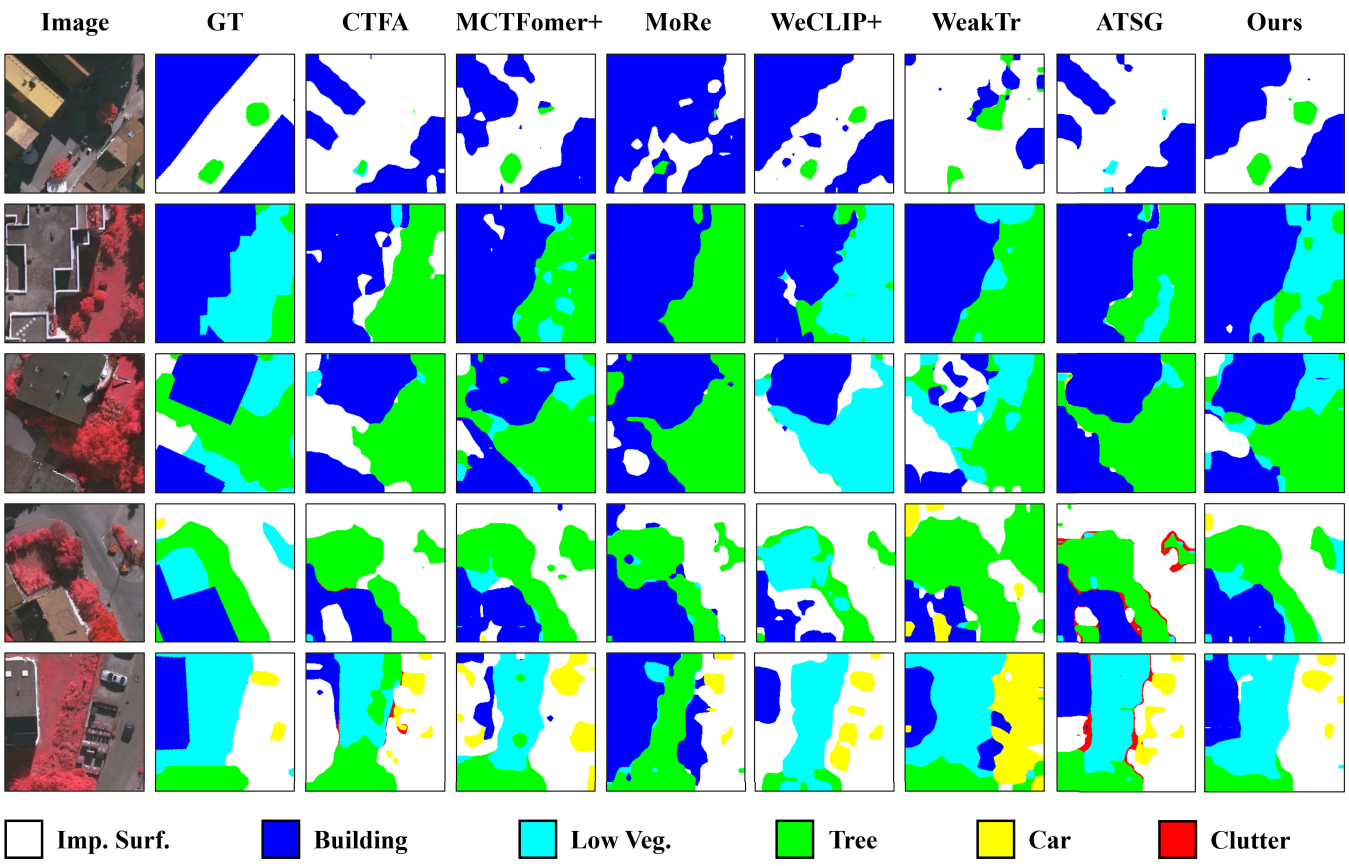


Fig. 3. Qualitative comparison of pseudolabels generated by different WSSS methods on the Vaihingen dataset.

for Rangeland, where GNATL reaches 36.97% IoU and exceeds MCTFormer+ by 12.15%. Although GNATL does not achieve the highest IoU for Urban and Water, its superior overall mIoU indicates stronger generalization across the six evaluated land-cover categories.

Fig. 4 presents qualitative comparisons on DeepGlobe. The proposed method produces more coherent land-cover regions and better preserves transitions among agriculture, rangeland, forest, and barren. Together with the Potsdam and Vaihingen results, these experiments show that GNATL generalizes from high-resolution urban aerial imagery to satellite imagery with more diverse geographical environments and land-cover distributions.

C. Ablation Studies

To verify the effectiveness of the proposed token learning modules, we conduct ablation studies on both the Potsdam and Vaihingen datasets.

1) Effect of NPTL: Table IV shows the effect of the proposed NPTL. On the Potsdam dataset, adding NPTL to the baseline improves the mIoU from 51.59% to 53.82%, bringing a gain of 2.23%. On the Vaihingen dataset, NPTL improves the mIoU from 42.97% to 47.21%, with a larger gain of 4.24%. These results demonstrate that enforcing consistency between overlapped neighboring crops can effectively constrain deep patch tokens and improve their local semantic discrimination.

When NPTL is combined with GCTL, the performance is further improved. Compared with the model using only GCTL, the full GNATL improves the mIoU by 2.66% on Potsdam and 2.55% on Vaihingen. This indicates that NPTL provides complementary patch-level constraints to the global class-token guidance, which helps alleviate patch token oversmoothing and improves segmentation performance.

2) Effect of GCTL: Table IV also verifies the effectiveness of the proposed GCTL. On the Potsdam dataset, introducing GCTL improves the baseline mIoU from 51.59% to 53.72%, resulting in a gain of 2.13%. On the Vaihingen dataset, GCTL improves the mIoU from 42.97% to 45.19%, with a gain of 2.22%. These improvements show that dynamically updated global class prototypes can provide stable semantic guidance for image-specific class tokens.

Moreover, when GCTL is added on top of NPTL, the full GNATL further improves the mIoU from 53.82% to 56.38% on Potsdam and from 47.21% to 47.74% on Vaihingen. This suggests that GCTL can reduce the semantic drift of image-specific class tokens and enhance the stability of category representations.

The quantitative and visualization results demonstrate that the proposed NPTL and GCTL modules are complementary. NPTL improves the local discrimination of patch tokens by imposing neighbor-aware consistency, while GCTL provides stable category-level guidance for image-specific class tokens.

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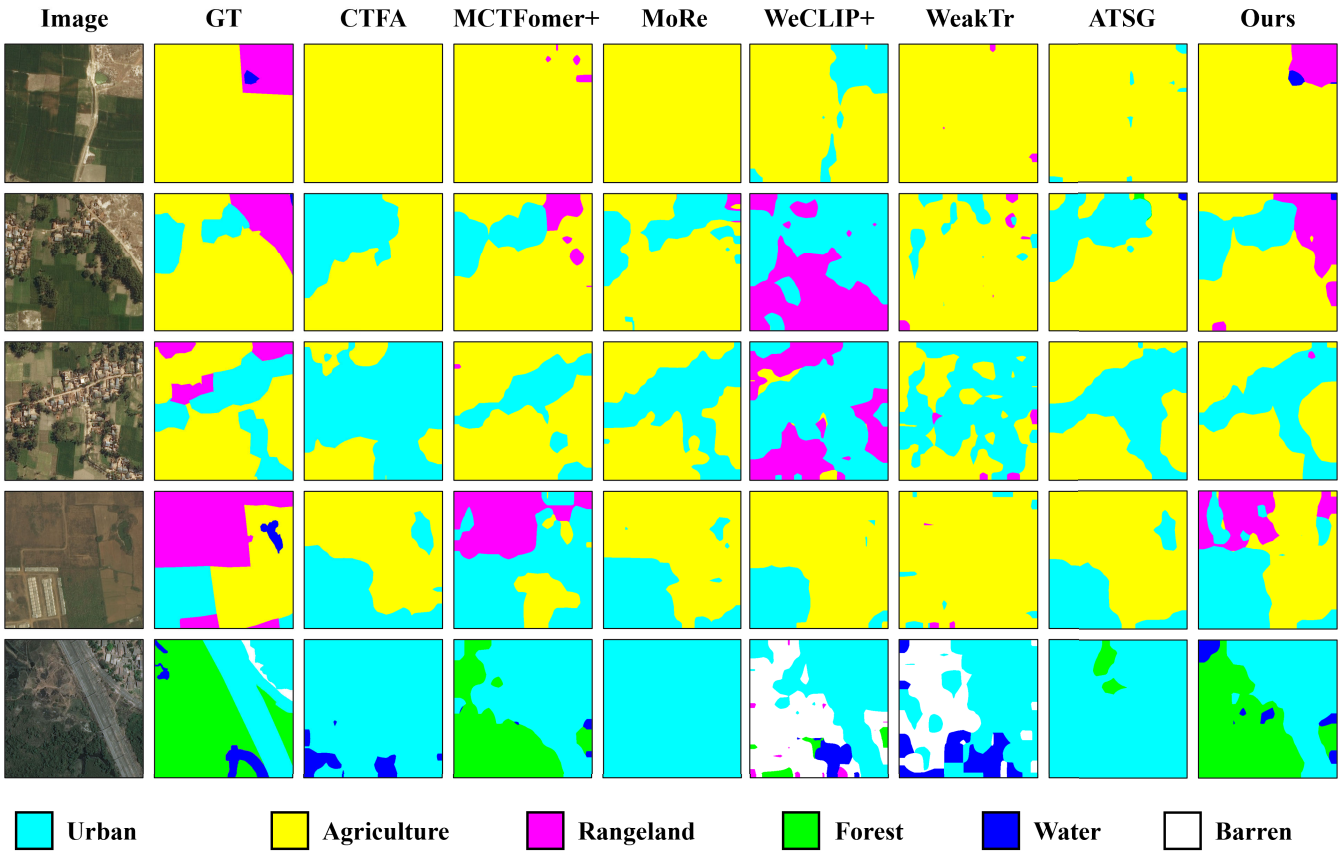


Fig. 4. Qualitative comparison of pseudolabels generated by different WSSS methods on the DeepGlobe dataset.

TABLE IV
ABLATION STUDY OF THE PROPOSED MODULES ON THE POTSDAM AND VAIHINGEN DATASETS

Dataset	Method	Imp. Surf. (%)	Building (%)	Low Veg. (%)	Tree (%)	Car (%)	Clutter (%)	mIoU (%)
Potsdam	Baseline	61.60	71.09	50.72	55.46	42.60	28.08	51.59
	Baseline + NPTL	65.17	75.28	56.16	57.17	40.15	28.97	53.82
	Baseline + GCTL	64.75	74.02	55.18	54.95	**45.70**	27.75	53.72
	Ours	**66.26**	**77.69**	**60.48**	**57.55**	42.48	**33.80**	**56.38**
Vaihingen	Baseline	53.94	58.64	39.07	57.91	21.78	26.48	42.97
	Baseline + NPTL	58.92	68.68	45.96	56.95	26.42	26.33	47.21
	Baseline + GCTL	51.69	67.55	42.19	**57.99**	18.55	**33.17**	45.19
	Ours	**59.70**	**69.01**	**46.65**	57.84	**27.24**	25.98	**47.74**

As a result, GNATL achieves more balanced performance across different land-cover categories.

D. Token-Level Analysis

To directly verify the patch token oversmoothing and class-token semantic drift discussed in our motivation, we conduct two token-level analyses on the Vaihingen dataset.

For each Transformer layer, we calculate the average pairwise cosine similarity among patch tokens. As shown in Table V, GNATL obtains lower similarity than the baseline in all 12 layers. The mean similarity decreases from 0.3754 to 0.3396. The difference is more evident in deeper layers: over Layers 7–12, the average similarity decreases from 0.4369 to 0.3877. These results provide direct evidence that NPTL prevents patch representations from becoming excessively homogeneous and preserves local semantic discrimination.

We further collect final-layer class tokens from images containing each category and calculate their cross-image variance. Table VI shows that GNATL reduces the variance in four of the six categories, including building, car, impervious surface, and tree. More importantly, the mean variance decreases from 8.164 to 7.129. Although clutter and low vegetation exhibit class-specific fluctuations, the lower overall variance indicates that GCTL generally produces more stable category-level representations and alleviates semantic drift across images.

E. Parameter Analysis

To analyze the influence of key hyperparameters in GNATL, we conduct parameter analysis on the Vaihingen dataset.

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TABLE V

LAYER-WISE AVERAGE PAIRWISE COSINE SIMILARITY OF PATCH TOKENS ON THE VAIHINGEN DATASET. LOWER VALUES INDICATE MORE DISCRIMINATIVE PATCH REPRESENTATIONS

Method	Layer 1	Layer 2	Layer 3	Layer 4	Layer 5	Layer 6	Layer 7	Layer 8	Layer 9	Layer 10	Layer 11	Layer 12	Mean
Baseline	0.4481	0.3192	0.2724	0.2592	0.2745	0.3105	0.3665	0.3890	0.4473	0.4588	0.4808	0.4789	0.3754
Ours	**0.4305**	**0.3137**	**0.2561**	**0.2413**	**0.2406**	**0.2671**	**0.3161**	**0.3422**	**0.4085**	**0.4061**	**0.4207**	**0.4323**	**0.3396**

TABLE VI

CLASS-WISE CROSS-IMAGE VARIANCE OF FINAL-LAYER CLASS TOKENS ON THE VAIHINGEN DATASET. LOWER VALUES INDICATE MORE STABLE CATEGORY-LEVEL REPRESENTATIONS

Method	Building	Car	Clutter	Imp. Surf.	Low Veg.	Tree	Mean
Baseline	9.910	13.574	**1.647**	10.604	**5.341**	7.906	8.164
Ours	**9.757**	**10.680**	2.402	**7.018**	6.637	**6.280**	**7.129**

TABLE VII

EFFECT OF THE NPTL LOSS WEIGHT λ_{nptl} ON THE VAIHINGEN DATASET

λ_{nptl}	Imp. Surf. (%)	Building (%)	Low Veg. (%)	Tree (%)	Car (%)	Clutter (%)	mIoU (%)
0.05	51.98	66.17	44.08	57.55	18.12	20.54	43.07
0.10	53.26	67.51	44.48	58.02	19.58	29.10	45.32
0.20	53.56	68.62	44.63	57.20	21.20	27.74	45.49
0.30	56.82	68.83	44.64	57.59	23.25	28.25	46.56
0.40	55.79	**69.50**	43.88	**58.84**	23.15	**31.85**	47.17
0.50	**59.70**	69.01	**46.65**	57.84	**27.24**	25.98	**47.74**
0.60	53.70	67.68	42.69	57.78	22.74	30.02	45.77

TABLE VIII

EFFECT OF THE SELECTED NPTL LAYERS ON THE VAIHINGEN DATASET

Layers	Imp. Surf. (%)	Building (%)	Low Veg. (%)	Tree (%)	Car (%)	Clutter (%)	mIoU (%)
Last 3	57.74	68.73	42.16	**58.99**	25.91	29.71	47.21
Last 6	**59.70**	**69.01**	**46.65**	57.84	**27.24**	25.98	**47.74**
Last 9	56.08	68.29	42.39	58.66	23.14	**30.04**	46.43

1) Effect of NPTL Loss Weight: Table VII shows the effect of the NPTL loss weight λ_{nptl}. The performance gradually improves when λ_{nptl} increases from 0.05 to 0.5, indicating that neighbor-aware patch consistency provides effective supervision for patch token learning. The best mIoU of 47.74% is achieved when $\lambda_{\text{nptl}} = 0.5$. When the weight is further increased to 0.6, the performance drops, suggesting that an overly strong consistency constraint may suppress the flexibility of token representations.

2) Effect of Selected NPTL Layers: Table VIII evaluates NPTL using the last 3, 6, and 9 Transformer layers. The last six layers achieve the highest mIoU of 47.74%. Using only the last three layers provides limited patch-level supervision, whereas extending the constraint to the last nine layers includes relatively shallow representations that mainly capture low-level appearance information and may overconstrain intermediate features. The last six layers therefore provide a better balance between semantic maturity and patch-level discriminability.

3) Effect of Neighboring-Crop Sampling Strategy: As shown in Table IX, random neighbor sampling achieves the highest mIoU of 47.74%. Fixed directional sampling restricts the spatial diversity of neighboring relations and can introduce directional bias. Random sampling exposes NPTL to more diverse spatial arrangements and is therefore adopted in all experiments.

TABLE IX

EFFECT OF THE NEIGHBORING-CROP SAMPLING STRATEGY ON THE VAIHINGEN DATASET

Strategy	Imp. Surf. (%)	Building (%)	Low Veg. (%)	Tree (%)	Car (%)	Clutter (%)	mIoU (%)
Top	**60.04**	**69.60**	45.91	55.87	25.79	23.84	46.84
Bottom	57.82	66.50	46.58	**61.16**	21.54	25.25	46.48
Left	55.61	68.52	45.98	59.35	21.14	18.24	44.80
Right	58.21	65.28	41.30	60.76	25.87	16.66	44.68
Random	59.70	69.01	**46.65**	57.84	**27.24**	**25.98**	**47.74**

TABLE X

EFFECT OF THE GCTL WARM-UP LENGTH ON THE VAIHINGEN DATASET

Epochs	Imp. Surf. (%)	Building (%)	Low Veg. (%)	Tree (%)	Car (%)	Clutter (%)	mIoU (%)
6	58.49	68.53	45.63	**58.73**	23.64	23.92	46.49
7	58.43	66.38	44.07	57.19	25.88	**26.79**	46.46
8	**59.70**	**69.01**	**46.65**	57.84	**27.24**	25.98	**47.74**
9	58.49	67.74	45.07	56.72	25.18	22.51	45.95
10	58.73	67.36	44.85	56.83	25.41	24.60	46.30

TABLE XI

EFFECT OF TOP-k SELECTION IN GCTL ON THE VAIHINGEN DATASET

k_g	Imp. Surf. (%)	Building (%)	Low Veg. (%)	Tree (%)	Car (%)	Clutter (%)	mIoU (%)
3	58.51	67.45	45.78	57.18	24.12	25.30	46.39
5	**59.70**	**69.01**	**46.65**	**57.84**	**27.24**	25.98	**47.74**
7	58.70	67.57	45.95	57.42	25.34	26.47	46.91
9	55.10	68.92	44.17	56.21	22.37	**29.10**	45.98

TABLE XII

EFFECT OF THE GLOBAL CLASS TOKEN UPDATE MOMENTUM μ ON THE VAIHINGEN DATASET

μ	Imp. Surf. (%)	Building (%)	Low Veg. (%)	Tree (%)	Car (%)	Clutter (%)	mIoU (%)
0.75	58.45	67.33	45.74	56.08	24.42	25.07	46.18
0.80	**59.70**	**69.01**	**46.65**	**57.84**	**27.24**	**25.98**	**47.74**
0.85	59.55	67.86	45.81	57.49	26.33	25.74	47.13
0.90	59.68	67.93	45.59	56.56	27.02	25.35	47.02

4) Effect of GCTL Warm-Up Length: Table X evaluates warm-up lengths from 6 to 10 epochs. The best result is obtained with an 8-epoch warm-up. Shorter warm-up periods activate GCTL before high-response patches become sufficiently reliable, while longer periods shorten the duration of global semantic guidance. Eight epochs provide an appropriate balance between prototype reliability and effective optimization.

5) Effect of Top-k Selection in GCTL: Table XI shows that selecting the top-5 high-response patches achieves the best mIoU of 47.74%. With only three patches, a prototype may not capture sufficient intraclass appearance variation. Increasing k_g to 7 or 9 introduces more relatively low-confidence patches and raises the risk of including irrelevant regions. Top-5 therefore provides a suitable balance between semantic reliability and intraclass diversity.

6) Effect of Global Class Token Update Momentum: Table XII reports the influence of the momentum coefficient

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TABLE XIII
EFFECT OF THE GCTL LOSS WEIGHT λ_{gctl} ON THE VAIHINGEN DATASET

λ_{gctl}	Imp. Surf. (%)	Building (%)	Low Veg. (%)	Tree (%)	Car (%)	Clutter (%)	mIoU (%)
0.10	**59.70**	**69.01**	**46.65**	**57.84**	**27.24**	25.98	**47.74**
0.20	59.32	66.54	45.53	57.02	26.15	24.94	46.58
0.30	57.40	66.18	46.11	56.45	23.52	23.06	45.45
0.40	56.71	65.35	45.94	55.69	22.68	**26.15**	45.42

TABLE XIV
COMPARISON OF MODEL EFFICIENCY WITH OTHER WSSS METHODS

Method	Params (M)	FLOPs (G)
MCTFormer+ [42]$_{TPAMI'24}$	**22**	**4.67**
CTFA [39]$_{TGRS'24}$	112	19.07
MoRe [40]$_{AAAI'25}$	95	19.07
WeCLIP+ [47]$_{TPAMI'25}$	175	24.02
WeakTr [37]$_{TIP'26}$	**22**	**4.67**
ATSG [41]$_{TGRS'26}$	277	19.07
Ours	**22**	**4.67**

μ for updating global class tokens. The best result is obtained when $\mu = 0.8$, achieving an mIoU of 47.74%. A smaller momentum may make global class tokens sensitive to batch-level variations, while a larger momentum may slow down the update process and weaken their adaptability. Therefore, μ is set to 0.8 in our experiments.

7) Effect of GCTL Loss Weight: Table XIII shows the effect of the GCTL loss weight λ_{gctl}. The model achieves the best mIoU of 47.74% when $\lambda_{gctl} = 0.1$. As the weight increases, the overall performance gradually decreases. This indicates that moderate global class-token guidance is beneficial, while an overly large alignment weight may reduce the adaptability of image-specific class tokens.

F. Efficiency Analysis

To evaluate the computational efficiency of the proposed GNATL, we compare the model complexity with representative WSSS methods. The comparison is conducted in terms of the number of parameters and FLOPs, as shown in Table XIV.

As shown in Table XIV, GNATL has the same model complexity as MCTFormer+ and WeakTr, with only 22M parameters and 4.67G FLOPs. This indicates that the proposed NPTL and GCTL modules do not introduce additional inference burden to the classification backbone. Compared with CTFA, MoRe, WeCLIP+, and ATSG, GNATL requires fewer parameters and lower computational cost.

Combined with the quantitative results in Tables I–III, GNATL achieves better pseudolabel quality while maintaining a lightweight model complexity. These results show that the improvement of GNATL mainly comes from more effective token learning rather than increasing model size or computational cost.

V. CONCLUSION

In this article, we have proposed GNATL for image-level weakly supervised remote sensing semantic segmentation. In particular, NPTL exploits the overlapped regions between neighboring crops to construct implicit patch-level supervision, which helps alleviate patch token oversmoothing and preserve local semantic discrimination. GCTL maintains dynamically updated global class tokens as category-level semantic prototypes, which provide stable semantic guidance for image-specific class tokens and reduce semantic drift. Experiments on the Potsdam, Vaihingen, and DeepGlobe datasets demonstrate that GNATL achieves better pseudolabel quality than representative weakly supervised segmentation methods while maintaining a lightweight model complexity. Token-level analyses further show that GNATL reduces mean patch token similarity by approximately 9.54% and mean cross-image class-token variance by approximately 12.68%, directly supporting the motivations of NPTL and GCTL.

Despite its effectiveness, GNATL has some limitations. First, the proposed NPTL relies on spatially overlapped neighboring crops, and its effectiveness may be affected by the crop size, stride, and the spatial distribution of land-cover categories. Second, GCTL updates global class tokens based on high-response patch tokens from mini-batches, which may still introduce noisy category information when activation maps are inaccurate in early training stages. In addition, small objects and categories with severe class imbalance remain challenging under image-level supervision.

In future work, we will further explore how to introduce complementary supervision signals to compensate for the limited localization information provided by image-level labels. One promising direction is to incorporate vision–language models, such as CLIP, to provide text-driven semantic supervision and improve category-level representation learning. Another direction is to exploit the inherent invariance properties of remote sensing images, such as scale invariance and rotation invariance, to design more effective regularization constraints.

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